

CHAPTER 2.2 - STORING SOLAR AND WIND: BATTERIES AND EFUELS

Fossil fuels were the "starter package" that nature "gifted" us to bootstrap our machine civilization. It's a quite extraordinary gift when you think about it. Four hundred million years before humans walked the earth, plants diligently started to convert solar energy into huge forests and swamps. Geological processes then buried them and turned them into coal deposits and gas fields. In the sea plankton cells did their part by turning solar energy into biomass and over similarly unimaginably long time scales they were converted into oil basins.

By the time humans wanted to kick off the industrial revolution, nature had everything ready for us: enormous reservoirs of compressed sunshine, conveniently stored, ready to be burned in our machines. Every year we are burning the equivalent of what nature captured for us in over a million years.

Now we are switching to a more permanent solution: directly harvesting sunshine using solar and wind. The good news is that solar and wind are cheap, abundant and last essentially forever. But the sun doesn't always shine and the wind doesn't always blow. So the bad news is that we need to create ways to store them.

Learning to leave fuel behind

After burning biomass for a million years, and creating a machine civilization build on burning fossil fuel, humans apparently have trouble letting go of the idea of burning fuel. That was almost tangible when I started to research this twenty years ago. Often it persisted in places where it clearly made no sense anymore when you argued from first principles.

It's the best explanation I have for the biofuels craze. Biofuels need roughly five hundred times more land than solar panels to drive a car one kilometre forward. That large area of land also needs to be fertile and directly competes with nature and food. From a system perspective it's obviously an incredibly bad idea. Worse than fossil fuels and much much worse than electric vehicles. Yet most policy makers and energy experts lacked that simple insight when I wrote my (Dutch) book on electric cars in 2009.

I think it's also the main reason people cling to nuclear and fossil fuel. Deep down they feel that energy supply that doesn't include storage simply cannot be relied upon, no matter how many fancy simulations with batteries and other forms of electric energy storage you show them. For me nothing else can explain the vehement attacks on prof. Mark Z Jacobson: in 2009 he was one of the first to propose an energy system without nuclear, fossil fuels or biofuels could work and ten years later many scientists could still not forgive him for this "heresy".

But it's becoming harder and harder to deny that solar, wind, batteries and e-fuels are becoming by far our cheapest option and keep developing like crazy. Meanwhile, the

competition -- fossil fuels, biofuels and nuclear fission -- is stuck. That means the playing field will continue to tilt towards renewables.

But how do we deal with periods without sunshine and wind?

Batteries to replace super-grids and space-based solar

A few decades ago, I didn't think batteries were the answer. I thought geometry was.

The sun, I thought, always shines somewhere on Earth. If you could just connect the place where the sun was shining to the place where the sun was needed, you would always have solar energy. Simple, right? When the sun set over Spain, it would be rising in the Caucasus. Same for wind. When the wind dropped over the Netherlands, it would be blowing over Morocco. With a sufficiently large and sufficiently low-loss super-grid, you would not need to store anything. When I started to research it in 2008, this seemed to be what everybody was thinking. I remember devouring every paper on superconducting cables I could find because that might become the holy grail of long distance electricity cables.

The continental-scale attempt was [DesertTec](#), an industrial consortium launched in Munich in October 2009 by Munich Re, Siemens, E.ON, ABB, Deutsche Bank and others. The proposed budget was 400 billion euros and the aim was [15% of European electricity by 2050](#), delivered north through new high-voltage direct-current cables under the Mediterranean. By 2013 the founding foundation had [walked out of its own consortium](#). The official autopsy named the Arab Spring, water scarcity in the Sahara, and inter-partner squabbles. In retrospect the real reason was probably that technology was moving towards distributed energy systems.

Another dead end I poured many many hours into was space based solar. A photovoltaic panel in geostationary orbit gets sunlight more or less constantly: no night, no clouds, no atmosphere in the way. It's not fringe science and far from impossible. [Peter Glaser at Arthur D. Little proposed it in 1968](#). The US Department of Energy [funded a major programme through the late 1970s](#). Japan's space agency ran an active research effort for decades. As recently as 2023, a team at [Caltech put a demonstrator into orbit](#) to test microwave power transmission from space. And with the cratering costs of space travel due to reusable rockets it is becoming cheaper every year. But it will be crazy expensive to start with, and you either need giant receivers on earth that are comparable in size to a land based solar array, or you have to content with concentrated death rays that could easily cook all humans in a city if something went wrong.

Now the solution is obvious of course. Batteries can easily overcome the day-night cycle and with a little e-fuel for seasonal storage we need neither worldwide super-grids, nor space-based solar. But back then, storage felt so intractable that beaming gigawatts of microwaves from a geostationary satellite felt like a more solvable problem than putting cheap batteries on the floor. We really have come a long way the last twenty years.

Understanding energy storage options

This chapter will talk mainly about battery storage, because they have the biggest and most underestimated impact. But it's important that you understand what other storage options

there are. Because batteries are not the optimal storage option in all situations. To understand why I created the table below.

Storage option	% losses	\$/kW	\$/kWh	kg/kWh	l/kWh	Typical duration
Diesel + motor	65	500	0.5	0.1	0.1	months–years
Corn ethanol + motor	70	600	1	0.15	0.2	months–years
Methanol from electricity	75	3,000	0.5	0.2	0.25	months–years
Ammonia from electricity	80	3,500	3	0.2	0.3	months–years
Hydrogen, 700 bar tank	65	2,500	20	0.6	1.2	days–months
Hydrogen, salt cavern	65	2,500	1	0.03	3	weeks–months
Pumped hydro (large existing)	22	2,000	15	700	700	hours–days
Thermal (molten salt)	60	700	30	45	25	hours–days
CAES (diabatic)	50	1,200	50	60	8	hours–days
Gravity storage (deep shaft)	18	1,200	300	370	50	hours
Flow battery (vanadium redox)	30	600	400	30	80	4–12 h
LFP battery pack	10	100	200	6	5	2–6 h
Flywheel	15	500	8,000	120	30	seconds–minutes
Supercapacitor	5	75	10,000	150	150	seconds

I've ordered them by the typical storage duration they are used for. Let's try to understand why a typical lithium iron phosphate (LFP) battery isn't always the best solution.

First, imagine you want to fly around the world. Your most important metric would be called "gravimetric energy density". Basically you want as few kilograms per kWh of energy as possible. Diesel has the same gravimetric energy density as jet fuel: 0.1 kg/kWh. LFP batteries weight 6 kg/kWh. That 60x more! It's clear that you cannot make a long distance plane with LFP batteries.

For short distance flying, batteries are fine, as you might have guessed from all the battery electric drones buzzing around these days. And batteries can become much lighter than current LFP. The best commercial batteries today are around 3.5 kg/kWh, lithium metal and solid state are expected to reach 2 kg/kWh within a decade, and with lithium sulfur (Li-S) 1 kg/kWh might be achievable. Take into account that electric engines are more efficient and battery powered flights of 2000 km would become attractive. Still, for transatlantic flights it's clear why most experts are betting on fuels, even in the medium to long term. A similar logic applies to long haul shipping, although off-shore wind farms equipped with fast chargers could plausibly solve that problem.

Seasonal storage is the other place where batteries find it hard to compete. Here the energy is only needed once or twice a year. The cost per kWh of storage becomes dominant in that

scenario. In that scenario it's cheaper to use electricity to produce hydrogen and put it in salt caverns, or to produce methanol. The cost per kWh of storage is simply much lower.

However, you would never do this for energy you want to store and retrieve every day or week. Why not? Look at the efficiency! If you convert electricity to hydrogen and then back to electricity again, you lose 65% of energy in the process. For LFP batteries you only lose 10%. In the long term, I expect sodium-ion battery packs to fall below \$20/kWh and cover durations up to two weeks.

Bottom-line: fuels produced with solar and wind, often called e-fuels, can take care of long duration storage and anything else fossil fuels can do. Their energy will be more expensive than the energy from batteries, but since only a tiny fraction of electricity will be turned into e-fuels, it does not have a big impact on overall system costs.

Supercapacitors are interesting at the other side of the spectrum. With graphene you could realistically go from 150 kg/kWh now to 5 kg/kWh in the long term. That's hardly heavier than lithium batteries now. But supercapacitors could do for robots and cars, what ATP does for human muscles: provide short bursts of power. And the burst of power supercapacitors could give a car or robot are mind boggling. Just one kWh of supercapacitor could theoretically deliver energy bursts of over 10 thousand kW.

So batteries will do the bulk of the work, but there are better alternatives for storing electricity over very long periods of time and over very short periods of time. If you are familiar with computers, a good analogy is data storage. Supercapacitors are comparable to fast DRAM memory, batteries to SSD hard drives and e-fuels to slow but large hard drives, glass or DNA storage.

Now let's get back to the main story: how to get from fossil fuels to solar and wind and how batteries help.

Demystifying batteries

In essence, a battery is just a place where positively charge atoms called ions can hang out. If you charge your battery, the ions leisurely swim towards the side of the battery that has the minus sign painted on it. We call this side the anode. This is the ions home where they chill before going to work. Their home is often made from a lattice of carbon atoms.

When you want the battery to do work for you, essentially what happens is that the ions swim to the other side of the battery with the plus on it that we call the cathode. Here they have their office. For a lithium iron phosphate battery the ions doing the work are lithium ions, and their office is made from iron-phosphate. For a sodium battery the workers are sodium ions and there are many different offices in development. From example offices from Prussian blue (iron, carbon and nitrogen) or from other abundant elements.

Building better batteries means learning how to build cheap but durable homes and offices for the ions and making sure they can get from home to work easily each day. There are also so called solid state batteries. Here the ions don't swim but walk to work because the electrolyte swimming pool between home and work has been replaced with a solid road.

As the ion workers swim or walk to their office, they are balanced out by electrons (negatively charged particles) flowing through the electricity wire of the equipment you want to power.

For over a hundred years, we used heavy batteries made from lead and acid. Somehow we never bothered to improve them. Then, with the rise of laptops and cell phones, we finally woke up and started to search for lighter ions. And there's still endless possibilities. Superlight batteries using lithium-sulphur or lithium-air. Super cheap batteries using sodium-sulphur or iron-air. Better electrolytes. Solid lithium/sodium/aluminium anodes. If you follow this space closely, you know that we have barely scratched the surface of what's possible.

Not a material problem

Many people are worried about the raw materials needed to make batteries. They shouldn't be. Just as the two or three hundred kilogram battery in an electric vehicle replaces more oil than the biggest oil truck can carry -- before the battery is recycled instead of burned -- stationary batteries are a spectacularly good addition to our energy system.

In part 1 chapter 10 we already saw that a solar system replaces 250 times its weight in coal, before it can be recycled twenty times or more. But what you might have missed is how little battery was needed in such a solar system. The glass of the solar panels alone weighted over four times more than the battery. And especially for stationary batteries the materials are unproblematic.

The materials are also potentially very cheap. [Keiner and colleagues](#) at LUT University, in February 2026, took a complete utility-scale lithium iron phosphate or sodium-ion system — cells, packs, racks, balance of plant, the lot — down to the raw materials and added the costs back up. Their medium scenario for 2050 produces a complete system at 29 to 52 dollars per kilowatt-hour. Their optimistic scenario lands between 11 and 14 dollars.

Learning to overtake even solar panels

In 2010, a kilowatt-hour of lithium-ion battery cost on average [about 1,474 US dollars in 2025 money](#), while by 2025 it was [108 dollars](#): a 93% drop in fifteen years. I wrote a paper with Floor Alkemade arguing the data indicates [roughly 28% per doubling \(Hoekstra & Alkemade\)](#). I think we can expect about 7 more doublings which would take the price to around 25 dollars per kWh. And if there's one thing I've learned over the past 20 years, it's not to bet against the learning curve.

Solar panels have been the poster child for fast learning in energy, but batteries are overtaking them. That's because manufacturing battery cells is an even purer example of producing small units and learning to improve that with each new product.

The installation of batteries is also relatively straightforward. Batteries need very little space and you don't need to bolt them onto your roof. Instead you can put them basically everywhere. They are already much safer and healthier than e.g. a gas stove while they get safer all the time so placing them indoors is increasingly unproblematic.

Batteries are also much easier to place than power lines or transformers. Power lines must (by definition) go from a defined point A to a defined point B and there's bound to be some areas that need to be crossed where there's natural challenges or people who are opposed. Transformers must (by definition) be at the crossroads between multiple power lines which constrains their position. Batteries must basically be somewhere in the neighbourhood of a power line. That's it. Placement is much less of a challenge and in the future they will provide resilience and decoupling everywhere. I think new houses and businesses without batteries will become exceptional within ten years.

Batteries to replace the local grid

That batteries help solar and wind is widely understood by now, but what few people understand yet is how they will transform the electricity grid. Especially the grid operators and their captured regulators seem incapable of seeing the obvious.

Grid operators mostly view batteries as a nuisance that can cause peaks on the grid when many batteries charge or discharge at the same time, not as a way to reduce peaks on the grids if you ask them nicely.

And yet, if the trend is clear anywhere its here: over the past 15 years the price of transformers has roughly quadrupled while over that same time, stationary batteries have become 15 times cheaper and 4 times longer lasting. So compared to transformers, battery cells have become almost a thousand times cheaper over the past 15 years. A thousand times. And yet the grid operators are acting as if nothing has happened and their old models don't need an update.

The cost of a grid connection depends on the size of the cable, the application, and where you need it. First size. Big cables are cheaper per kWh they transport. So if you have a lot of cheap energy in one place, and a lot of demand in the other, electricity cables will probably always be a good idea. Then application. If peaks and troughs in production are long lasting, batteries get more expensive the longer they last. That is why batteries are a much better complement for solar (with its day-night cycle) than wind (where fluctuations can last many weeks). Finally where you need it. If buildings are close together, connecting all of them with using an electricity grid is much more cost effective than when they are far apart.

So... for a dense city with little space for solar and a lot of wind nearby, going off-grid will probably remain unattractive. But for a rural area with a lot of solar, grids are already not attractive anymore. And as batteries get cheaper and local production with solar increases, batteries will keep chipping away at the need for a grid connection.

Even in cities that need a grid, batteries can optimise the local use of energy while reducing the peaks that determine the size of the cable and transformer you need. With my company I just researched this in a study for the Dutch government and we found that smart charging electric vehicles and using small batteries of 5 kWh per house could roughly halve the collective peak houses cause on a neighbourhood transformer. For the Netherlands where grid congestion is costing society over 15 billion dollar per year this could save billions per year. Worldwide we are talking about an upside of hundreds of billions every year.

And cost is only the beginning. A grid built around distributed storage, with batteries scattered through cars and homes and substations can be more resilient. Each part can keep functioning when other parts have failed. Each part has a degree of autonomy in deciding when to charge and discharge that no centralised generator could match. Each part is small enough that any individual failure is local. Such a system is far better able to gracefully deal with really big shocks -- hurricanes, cyber-attacks, wars and a closed street of Hormuz -- than the centralised grid that came before it.

But, I'm getting ahead of myself. Before we talk more about our resilient future with batteries, the next chapters will introduce you to two battery companions. They are revolutionary on their own, but together with solar the four of them are completely unstoppable.